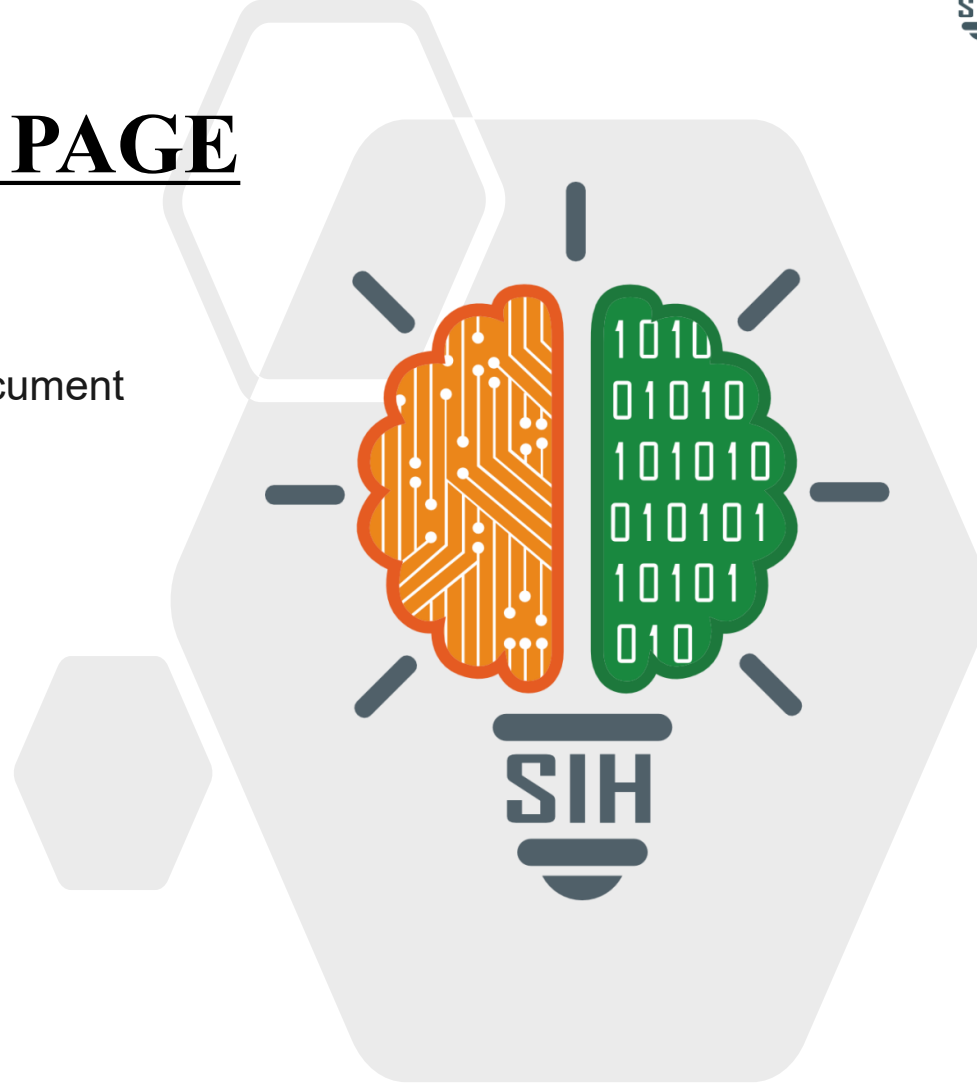


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID –** 26188
- **Problem Statement Title-** AI-Based Fake Identity & Document Screening System
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-** 156702
- **Team Name (Registered on portal)-** Digital Warriors



AI-Based Fake Identity & Document Screening System

❖ Proposed Solution:

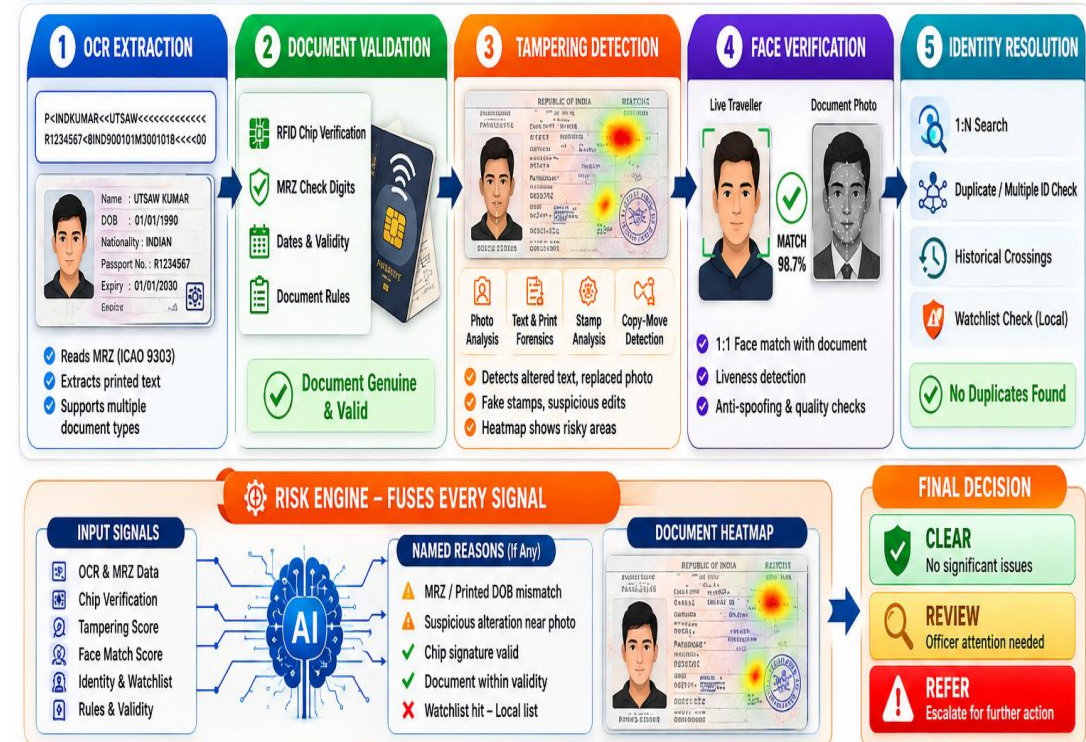
- OCR, MRZ, barcode, and chip-data validation
- Detects forged, altered, and tampered documents
- Cross-checks inconsistencies across document elements
- Provides explainable fraud alerts
- Generates an authenticity/risk score !!

❖ How it Address the Problem:

- Use **AI + OCR** to extract and analyze document information'
- Verify **MRZ, QR/barcodes, and chip data** for consistency
- Cross-check identity details across multiple data sources
- Provide **explainable alerts** and a **final risk score** for human verification!!

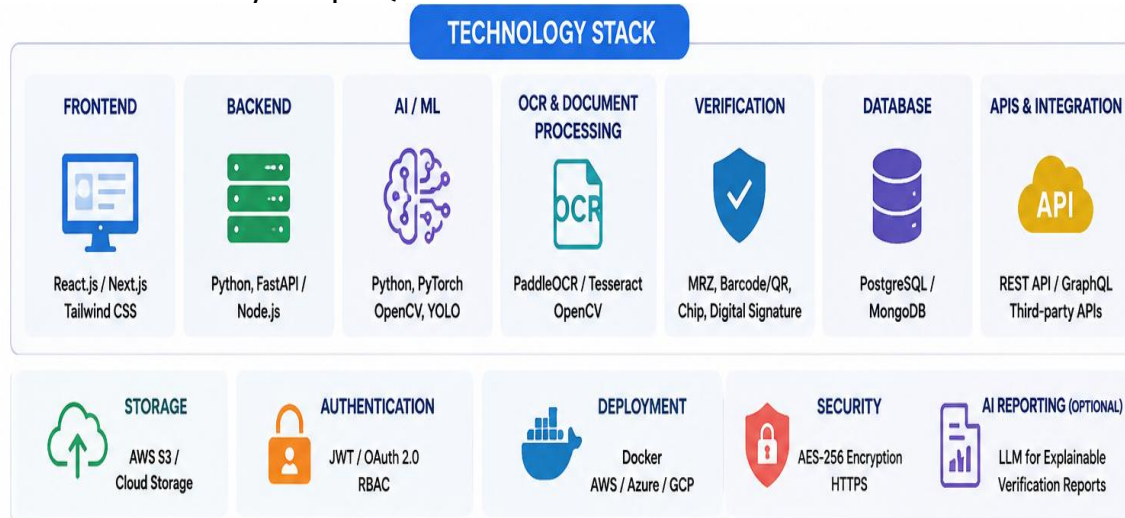
❖ Innovation and Uniqueness:

- **Proof-based verification** instead of only probability-based AI
- **Contradiction detection** to identify mismatches as fraud signals
- **Zero-database forgery detection** for previously unseen fake documents
- **Explainable AI alerts** showing exactly what failed and where
- **Deepfake-resistant face verification** between document photo and live person!!



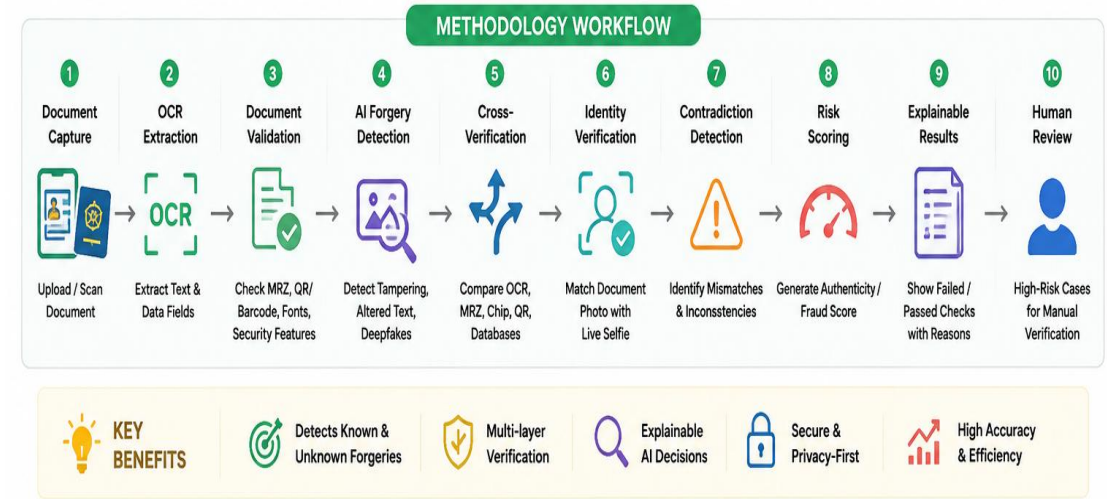
❖ Technologies to be used:

- **Frontend:** React.js / Next.js + Tailwind CSS
- **Backend:** Python FastAPI / Node.js
- **AI/ML:** Python, PyTorch, OpenCV, YOLO, Hugging Face Transformers
- **OCR:** PaddleOCR / Tesseract / Google Cloud Vision API
- **Document Analysis:** OpenCV + deep-learning models for tampering detection
- **Barcode/QR:** ZXing / OpenCV
- **Digital Verification:** ICAO ePassport/PKI and NFC chip verification
- **Database:** PostgreSQL / MongoDB
- **Authentication:** JWT + OAuth 2.0
- **APIs:** REST API / GraphQL



❖ Methodology and process for implementation:

- **Document Capture** → Upload/scan ID document and capture image.
- **OCR Extraction** → Extract name, DOB, ID number, and other fields.
- **Document Validation** → Check MRZ, QR/barcode, fonts, layout, and security features.
- **AI Forgery Detection** → Detect image manipulation, altered text, photos, and deepfakes.
- **Cross-Verification** → Compare information across OCR, MRZ, chip, barcode, and databases.
- **Identity Verification** → Match the document photo with a live selfie using face recognition.
- **Human Review** → Send high-risk cases for manual verification.



❖ Analysis of the feasibility of the idea:

- **Technically feasible** using existing AI, OCR, computer vision, and document-verification technologies.
- **AI models** can detect document tampering, fake images, altered text, and face mismatches.
- **APIs and databases** can support identity and document cross-verification.

Potential challenges and risks:

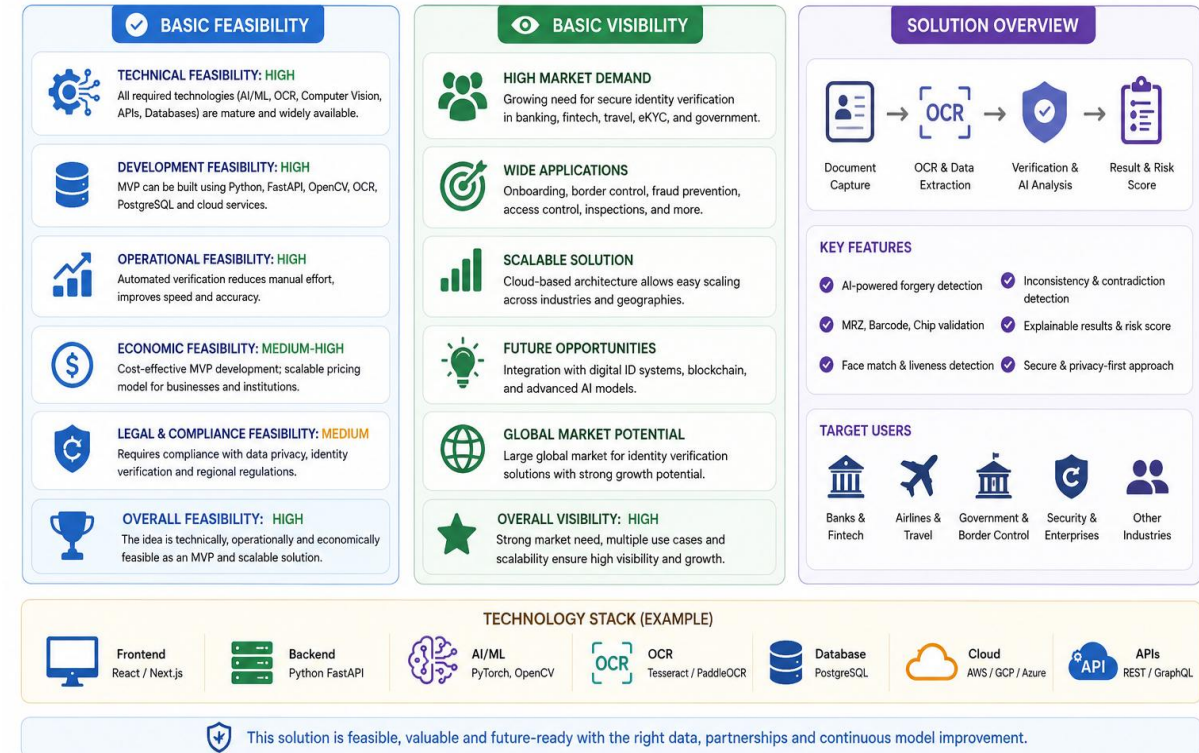
- Document diversity
- AI accuracy
- High-quality forgeries
- Data privacy
- Database/API access
- Spoofing attacks
- Security threats
- Model bias
- Processing cost
- Legal compliance

❖ Strategies for overcoming them:

- Document diversity
- AI accuracy
- Advanced forgeries
- False results
- Data privacy
- Limited APIs/databases
- Deepfakes/spoofing →
- Security threats
- Model bias
- Legal risks

AI-BASED FAKE IDENTITY & DOCUMENT VERIFICATION SYSTEM

Feasibility at a Glance | Visibility for the Future



< 3 sec

per document, versus
several minutes manually

4

independent face comparisons
per traveller

100%

of decisions logged with a
signed, tamper-evident trail

❖ Potential impact on the target audience:

- **Border officers (SSB, Bureau of Immigration)** — screening collapses from minutes to seconds, and the decision rests on evidence rather than intuition at the end of a long shift
- **Travellers** — genuine documents clear quickly and queues shorten; anyone flagged gets named reasons and a documented route to redress
- **Investigators and prosecutors** — a signed, hash-chained record per crossing, with a Section 63 electronic-evidence certificate generated automatically
- **Ministry of Home Affairs** — the same forgery scores the same at every checkpoint, replacing checkpoint-by-checkpoint variation with a national standard

❖ Benefits of the solution:

- **Security** — detects tampering and impersonation that no watchlist contains, including morphed passport photographs designed to match two people
- **Social** — consistent, non-arbitrary treatment at the counter; error rates measured and published by demographic group rather than left unexamined
- **Economic** — a force multiplier — higher throughput without additional posts, on commodity hardware instead of per-seat commercial SDK licensing
- **Operational** — offline-first, so it keeps working at low-connectivity frontier posts where a cloud service simply would not
- **Environmental** — edge inference avoids datacentre load, and a digital evidence trail replaces paper case files and repeated physical re-inspection
- **Institutional** — forgery patterns accumulate into searchable intelligence, turning every crossing into data for future detection

✓ Details / Links of the reference and research work:

❖ Standards and specifications

- ICAO Doc 9303, Parts 3, 4, 10, 11 and 12 — machine readable travel documents, logical data structure and PKI (icao.int)
- ICAO Public Key Directory — CSCA and document signer trust anchors (icao.int/Security/FAL/PKD)
- ISO/IEC 30107-3 — presentation attack detection reporting; ISO/IEC 19794-5 and 39794-5 — face image data

❖ Models and methods

- TruFor: Leveraging All-Round Clues for Trustworthy Image Forgery Detection and Localization — CVPR 2023
- CAT-Net: Compression Artifact Tracing Network for splicing localisation — IJCV 2022
- MVSS-Net++: multi-view multi-scale supervision for image manipulation detection — TPAMI 2022
- AdaFace: Quality Adaptive Margin for Face Recognition — CVPR 2022; ArcFace — CVPR 2019
- NIST FRVT Part 4 (MORPH) — morphing attack detection benchmark (nist.gov)

❖ Datasets

- MIDV-500 / MIDV-2019 / MIDV-2020 — identity document images and video, Institute for Systems Analysis, RAS
- MIDV-Holo — holographic security feature forgery detection
- DocXPand-25k and SIDTD — synthetic and tampered identity documents
- DocTammer (2023) — document-specific tampering localisation
- FRLL-Morphs, SMDD — face morphing benchmarks; CASIA-SURF, OULU-NPU — presentation attack
- CASIA v2, IMD2020, DEFACTO, CoMoFoD — general image manipulation

❖ Indian systems and data sources

- API Setu — NIC platform for government issuer APIs including Sarathi and Vahan (apisetu.gov.in)
- DigiLocker Partner APIs — issued document retrieval with citizen consent (partners.digitallocker.gov.in)
- UIDAI — Aadhaar secure QR code and paperless offline e-KYC specifications (uidai.gov.in)